

Recall, the Drinfeld curve is

$$Y = \{xy^q - yx^q = 1\} \subseteq \mathbb{A}^2(\mathbb{F})$$

$G = \mathrm{SL}_2(\mathbb{F}_q)$ acts by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} (x, y) = (ax+by, cx+dy)$$

μ_{q+1} acts by $\alpha(x, y) = (\alpha x, \alpha y)$. We showed last time the action of G on Y is free.

Prop The action of μ_{q+1} on Y is free.

Rmk The action of $G \times \mu_{q+1}$ is not free. For eg. $(-I_2, -1) \in G \times \mu_{q+1}$ acts trivially. In fact the action of $G \times \mu_{q+1} / \{\pm(I_2, 1)\}$ is still not free.

We will want to consider various quotients of Y by subgroups of $G \times \mu_{q+1}$. First, a definition

Def Let k be a field, X a k -variety, and Γ a finite group acting on X (on the left) by k -variety automorphisms. A quotient of X by Γ , if it exists, is a morph of k -varieties

$$\pi: X \rightarrow Z$$

s.t. 1. $\pi \circ g = \pi \quad \forall g \in \Gamma$

2. π is universal wrt 1, i.e. for any k -variety W and morph $f: X \rightarrow W$ satisfying $f \circ g = f \quad \forall g \in \Gamma$, $\exists$ unique morphism $h: Z \rightarrow W$ s.t. $f = h \circ \pi$.

Rmk If it exists, it is unique up to unique isomorphism, and we denote it $\Gamma \backslash X$.

Thm If X is quasi-projective, then $\Gamma \backslash X$ exists.

Sketch when X is affine

We have an equivalence of categories

$$\{\text{affine } k\text{-varieties}\} \longleftrightarrow \{\text{reduced fin type } k\text{-algs}\}^{\text{op}}$$

$$\mathrm{MaxSpec} A \longleftrightarrow A$$

So if $X = \mathrm{MaxSpec} A$, $A \ni \Gamma$ by k -alg auts.

Then $A^\Gamma := \{a \in A: ag = a \quad \forall g \in \Gamma\}$ a sub k -alg of A .

It is reduced since $A^\Gamma \subseteq A$ and A is reduced.

Let's show it is fin type $/k$.

First $A^\Gamma \subseteq A$ is integral since any $a \in A$ is a root of

$$\prod_{g \in \Gamma} (x - ag) \in A^\Gamma[x].$$

Since A is fin gen $/k$, we can find a fin gen k -alg $B \subseteq A^\Gamma$ s.t. A is integral over B . Thus A is a fin gen integral ext of B .
 $\Rightarrow A$ is a fin B -mod

B is Noeth since it is fin gen $/k$, so $A^\Gamma \subseteq A$ is a fin gen B -mod. Then A^Γ is fin type $/k$.

So $Z := \mathrm{MaxSpec} A^\Gamma$ is an affine k -variety. And the above argument shows $A^\Gamma \rightarrow A$ is finite and injective, so

$$\pi: X \rightarrow Z \text{ is finite and surjective.}$$

If $P \in X = \mathrm{MaxSpec} A$, and $g \in \Gamma$, clearly
 $P \mapsto P \cap A^\Gamma$
 $Pg \cap A^\Gamma = P \cap A^\Gamma$

So $\pi \circ g = \pi$ for all $g \in \Gamma$.

And if $W = \mathrm{MaxSpec} R$ is an affine k -var with $f: X \rightarrow W$ s.t. $f \circ g = f$ for all $g \in \Gamma$,

then f is induced by

$$\phi: R \rightarrow A \text{ satisfying } g \circ \phi = \phi \quad \forall g \in \Gamma$$

$\Rightarrow \mathrm{im}(\phi) \subseteq A^\Gamma$, so ϕ factors uniquely through

$$A^\Gamma \subseteq A$$

$\Rightarrow f$ factors uniquely through $\pi: \mathrm{MaxSpec} A \rightarrow \mathrm{MaxSpec} A^\Gamma$. $\square$

Rmk To prove the quasiprojective version, you glue the above construction. A subtle point is that to do so, you need to know that for every $x \in X$, the Γ -orbit $\Gamma \cdot x$ is contained in an open affine. To check this in the quasiproj case, embed $X \hookrightarrow \mathbb{P}^n$. Then for any fin collection of pts in X , can construct a homog polynomial that is nonzero on all these pts, the complement of its 0-locus will be open affine.

Prop Let $\pi: X \rightarrow Z$ is a morph of smooth irred k -varieties, and let Γ be a finite group acting on X . Assume

1. π is surjective.

2. $\pi(x) = \pi(y) \iff \Gamma x = \Gamma y$

3. $\exists x_0 \in X$ s.t. the differential of π at x_0 is surjective.

Then $\pi: X \rightarrow Z$ is the quotient of X by Γ .

Quotients of the Drinfeld curve

$$Y = \{xy^q - yx^q = 1\} \subseteq \mathbb{A}^2(\mathbb{F}) \hookrightarrow G \times \mu_{q+1}$$

Define

$$\gamma: Y \rightarrow \mathbb{A}^1(\mathbb{F})$$

$$(x, y) \mapsto xy^{q^2} - yx^{q^2}$$

This satisfies

$$\begin{aligned} \bullet \text{ For } \alpha \in \mu_{q+1}, \quad & \gamma(\alpha(x, y)) = \gamma(\alpha x, \alpha y) \\ & = \alpha^{q^2+1} \gamma(x, y) = \alpha^2 \gamma(x, y) \end{aligned}$$

$\bullet \gamma$ is $\mathbb{F}$ equiv.

$$\bullet \text{ For } g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G,$$

$$\begin{aligned} \gamma(g(x, y)) &= \gamma(ax+by, cx+dy)^{q^2} \\ &= (ax+by)(cx+dy)^{q^2} - (cx+dy)(ax+by)^{q^2} \\ &= (ax+by)(cx^{q^2}+d)^{q^2} - (cx+dy)(ax^{q^2}+by^{q^2}) \\ &= (ad-bc)xy^{q^2} - (ad-bc)yx^{q^2} \\ &= \gamma(x, y) \end{aligned}$$

Prop $\gamma: Y \rightarrow \mathbb{A}^1(\mathbb{F})$ is a quotient of Y by G .